

CE 773: Geometric Design and Analysis of High-Speed Roadways

Compiled Coursework Assignments 1–4

CE 773 Coursework Assignments: Geometric Design and Analysis of High-Speed Roadways



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Spring 2025

Abstract

This document compiles four coursework assignments submitted for CE 773: Geometric Design and Analysis of High-Speed Roadways, at the Indian Institute of Technology Bombay. Assignment 1 reviews the geometric design concepts applied in the Agra–Lucknow Expressway project. Assignment 2 estimates the Average Daily Traffic (ADT) and 30th-hour design volume for a rural four-lane highway from 50 days of hourly traffic-count data, and evaluates the Level of Service (LOS) over a 10-year design horizon to assess whether a 4-to-6 lane capacity expansion is warranted. Assignment 3 develops the tangent cross-section of a four-lane expressway located in the high-rainfall North-East region of India, based on IRC design guidelines. Assignment 4 is a field-based geometric design and safety review of the H10 intersection on the IIT Bombay campus. Each assignment is reproduced here as submitted, organised as one section per assignment, with the original questions, calculations, tables, and figures preserved. Assignment 1 was a group submission completed jointly with Shivam Singh (Roll No. 24M0552), and Assignment 4 was a group submission completed jointly with Karshan M. Kanjariya (24D0322), Irfan Ali (24D0314), Menbere Aklilu (24D0315), and Fred Ssemwogerere (24D0326); this is noted explicitly in the respective sections below. Assignments 2 and 3 were solo submissions.

Keywords: Highway Geometric Design; Traffic Volume Analysis; Level of Service; Expressway Cross-Section; Intersection Safety Review; IRC Design Standards

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1 Assignment 1: Case Study Review – Agra–Lucknow Expressway

Group submission note: This assignment was completed jointly with Shivam Singh (Roll No. 24M0552), as recorded on the original submission. The content below reproduces the submitted group report.

Question. Review any significant highway development project implemented in the last decade and describe the different design concepts applied.

1.1 Project Overview

The Agra–Lucknow Expressway, a 302.22 km access-controlled, six-lane expressway in Uttar Pradesh, was reviewed to evaluate its geometric design concepts and how it addresses challenges in transportation, safety, and sustainability. The expressway was developed to tackle transportation issues in Uttar Pradesh, reducing travel time from over 6 hours to approximately 3.5 hours between Agra and Lucknow, addressing congestion and safety concerns on NH-19, and integrating districts including Kannauj, Auraiya, and Unnao.

Table 1: Agra–Lucknow Expressway: project overview.

Feature	Details
Length	302.22 km
Lanes	6 lanes, expandable to 8 lanes
Construction period	January 2015 to November 2016
Completion time	22 months
Inauguration date	21 November 2016
Travel time reduction	From 6 hours to approximately 3.5 hours
Starting point	Agra Inner Ring Road
Ending point	State Highway 40 in Lucknow
Districts covered	Agra, Firozabad, Mainpuri, Etawah, Auraiya, Kannauj, Kanpur Nagar, Hardoi, Unnao, Lucknow

Table 2: Cost overview.

Aspect	Details
Initial estimated cost	Rs. 15,000 crore
Final completion cost	Rs. 13,200 crore
Cost savings	Rs. 1,800 crore

Table 3: Major stakeholders.

Stakeholder	Role
Government of Uttar Pradesh	Project financing and oversight
UP Expressways Industrial Development Authority (UPEIDA)	Project implementation and management
Construction contractors	Various firms responsible for different sections of the expressway
Local communities	Benefit from improved connectivity and economic opportunities

1.2 Design Concepts Applied

1.2.1 Greenfield Alignment Design

The expressway was built to solve problems with the busy NH-19 road, which faced heavy traffic, delays, and safety concerns. Expanding the old road would have needed much land in crowded areas, causing more displacement and higher costs. The new expressway takes a shorter route around cities, allowing faster and smoother travel; advanced technologies such as GIS and LiDAR were used in its planning.

1.2.2 Complete Street Concepts

The Agra–Lucknow Expressway does not incorporate complete street concepts to accommodate all users, including pedestrians, cyclists, motorists, and transit vehicles. It is designed only for fast-moving vehicles and does not have bike lanes or sidewalks for non-motorized users; it is focused on high-speed travel rather than mixed-use transportation. However, side roads for local access allow residents to reach their destinations.

1.2.3 Median and Roadside Safety Design

The expressway has wide medians (10–12 m) and crash barriers to help prevent head-on collisions and reduce the severity of accidents. The landscaped medians also improve the aesthetic and environmental character of the corridor, balancing safety with visual appeal.

1.2.4 Context-Sensitive Solutions (CSS)

The expressway aligns well with context-sensitive solutions by addressing the region’s specific needs while minimising environmental and social impact. The alignment was carefully planned to avoid urban centres, reducing disruptions in densely populated areas. The project also integrates green infrastructure, such as green belts along the expressway, noise barriers to reduce sound pollution, and rainwater harvesting pits every 500 m to address water management.

1.2.5 Performance-Based Design

The expressway's performance-based design ensures safety and operational efficiency. Grade-separated intersections remove level crossings, improving traffic flow and reducing delays; flyovers and underpasses are placed at key crossroads, while trumpet and cloverleaf interchange designs enhance connectivity and minimise conflict points. Intelligent Transport Systems (ITS), including surveillance cameras, electronic call boxes, digital message boards, and incident detection systems, further enhance safety and traffic management.

1.2.6 Practical Design

The expressway adopts a practical design approach, focusing on essential features to balance cost and functionality: limited interchanges for access control, service roads for local traffic, rumble strips, and reflective signage. This approach enabled the expressway to be completed within budget while meeting high operational standards.

Emergency airstrip integration. A standout feature of the expressway is its 3.3 km-long emergency airstrip near Unnao, designed to support disaster management and defense operations. Before this project, India lacked infrastructure capable of handling emergency fighter-jet landings on highways; the airstrip supports Indian Air Force (IAF) and disaster relief operations.

1.2.7 Design Matrix Approach

The design matrix approach was not explicitly applied during the planning and construction phases of the expressway. While this method typically involves choosing between basic, modified, or complete design standards based on project goals, the expressway adheres to high-speed geometric standards suitable for its purpose without a formal design matrix.

1.2.8 Safe System Approach

The expressway prioritises safety with well-designed curves, wide medians, crash barriers, and a 120 km/h speed limit. ITS enhances safety through prompt incident detection and response; green (solar) energy is used for all illumination; staggered service roads are provided on both sides of the expressway except at major bridges and RoBs; and patrolling vehicles and GPS-based ambulance services ensure rapid assistance within 4–11 minutes.

1.2.9 Travel Time Reliability Approach

The travel time reliability approach has improved consistency and predictability by eliminating traffic signals, restricting access, and integrating real-time traffic monitoring, reducing travel time between Agra and Lucknow to about 3.5 hours.

1.2.10 Value Engineering

The expressway successfully utilised value engineering to optimise costs without compromising quality. Through efficient land acquisition, streamlined construction processes, and practical design practices, the project achieved cost savings of Rs. 1,800 crore, reducing total expenditure from an initial estimate of Rs. 15,000 crore to Rs. 13,200 crore.

1.2.11 Designing of 3R Projects

The Agra–Lucknow Expressway is a greenfield project built from scratch and does not fall under 3R projects (Resurfacing, Restoration, Rehabilitation). Unlike 3R projects that upgrade existing infrastructure, it is a new access-controlled highway designed for high-speed connectivity without links to older roads.

1.2.12 Performance-Based Practical Design (PBPD)

The expressway is designed for good performance while keeping costs down. It starts with six lanes but can be expanded to eight if needed, includes technology for traffic management, and incorporates green spaces for sustainability – a design focused on safety, usability, and budget that can adapt to future needs.

2 Assignment 2: ADT, 30th-Hour Design Volume, and LOS Forecast for a Rural Four-Lane Highway

Question 1. In a rural median-divided four-lane highway development project, find the design traffic volume. The hourly traffic volume (in PCU) for 50 days per calendar year for one direction is provided in a spreadsheet keyed to the student's roll number. Estimate the ADT and the 30th-hour design volume, showing calculations step by step.

2.1 Given Data

- Rural, median-divided, four-lane highway.
- Hourly traffic volume (in PCU) for 50 days per calendar year, one direction.

2.2 Step-by-Step Solution

(i) **Estimation of ADT.** The Average Daily Traffic (ADT) is the total volume during a given period (greater than one day, less than one year), divided by the number of days in that period. Here, time period = 50 days; sum of daily traffic (for 50 days) = 1,638,192.

$$\text{ADT} = \frac{1,638,192}{50} = 32,763.84 \text{ PCU/day/lane}$$

Inference: the average number of vehicles passing the section per day over the observed period was approximately 32,763 PCU per day per lane.

(ii) **Estimation of 30th-hour design volume.**

- *Step 1.* Combine all traffic data for 50 days into one column, giving $50 \times 24 = 1200$ hourly rows.
- *Step 2.* Sort the 1200 values in descending order.
- *Step 3.* Identify the 30th highest value – the value exceeded 29 times.

Result: the 30th-hour design volume = **2157 PCU/hr/lane**.

Figure 1 shows the hourly-volume curve (hourly two-way traffic volume as a percentage of ADT, plotted against the number of hours in one year with volume greater than that shown), with the 30th and 100th highest-hour markers indicated. As noted on the original chart, this curve comes out as a straight line because the dataset covers only 50 days rather than a full year, which affects the design-volume consideration. Since there is no sizeable seasonal variation apparent in the data, the 30th highest volume is taken as 15% of ADT.

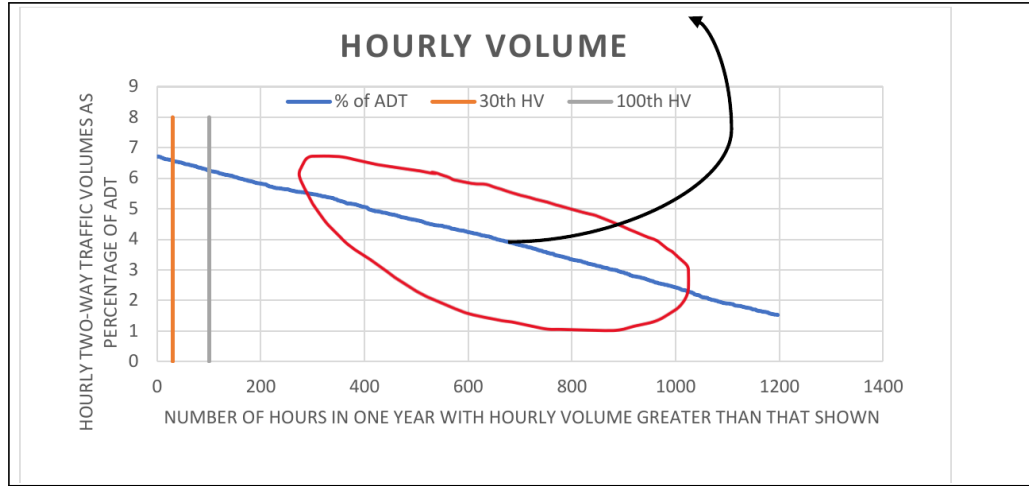


Figure 1: Hourly two-way traffic volumes as a percentage of ADT, showing the 30th and 100th highest-hour markers (as plotted in the original submission).

Question 2. The operating speed of the highway in Question 1 is 100 km/h. Using the estimated design volume, review the design improvement required over the next 10 years. The expected LOS at any point should not fall below “C.” Any improvement takes about 3 years for design, ROW acquisition, and construction. Show the evaluation and decision-making process step by step.

2.3 Given Data and Constraints

- Operating speed = 100 km/h; capacity for 4 lanes = 4540 PCU/hr per 2-lane carriageway; capacity for 6-lane divided highways = 6790 PCU/hr per 3-lane carriageway.
- Growth rate = 5% (per IRC).
- Expected LOS at any point should not fall below “C.”
- Design, ROW acquisition, and construction take about 3 years.

2.4 Step 1: Analysis of the Present Scenario Using the 30th-Hour Design Volume

Using the present design volume of 2157 PCU/hr/lane against a capacity of 4540 PCU/hr/lane, and projecting forward at a 5% annual growth rate using

$$\text{Future Volume} = \text{Current Volume} \times (1 + \text{Growth Rate})^{\text{Years}},$$

the future volume after 10 years is $2157 \times (1.05)^{10} = 3513.53$ PCU/hr/lane.

Table 4: 4-lane divided highway: design volume, v/c ratio, and LOS over 10 years, based on the 30th-hour design volume.

Year	Current	1	2	3	4	5	6	7	8	9	10
Design volume	2157	2264.85	2378.09	2497.00	2621.85	2752.94	2890.59	3035.12	3186.87	3346.21	3513.53
Capacity	4540	4540	4540	4540	4540	4540	4540	4540	4540	4540	4540
v/c	0.4751	0.4989	0.5238	0.5501	0.5775	0.6064	0.6367	0.6685	0.7020	0.7371	0.7739
LOS	C	C	D	D	D	D	D	E	E	E	E

The 4-lane highway maintains LOS C only until year 2, after which it drops to LOS D and eventually LOS E. Since LOS begins dropping after the 1st year, using the 30th-hour design volume directly does not leave enough time (3 years are required) to complete a capacity-improvement construction before the LOS constraint is violated. The design volume therefore needs to be reconsidered.

2.5 Step 2: Selecting a More Suitable Design Volume

Per IRC guidance on Design Hourly Volume for rural four-lane highways, design typically assumes moderate traffic growth over the next 10–20 years, and the 95th-percentile volume is sometimes used to account for more extreme future traffic conditions. Analysing the frequency distribution of the one-lane traffic volumes (Figure 2) shows that most traffic volumes (266, 270, 299, 294 hours in the respective bins) fall in the lower ranges, between 500 and 2100 PCU, while the (2100, 2500] range has the least volume (71 hours), indicating that high traffic volumes in this range are rare. The roadway therefore experiences moderate traffic conditions, with infrequent occurrences of exceptionally high volumes.

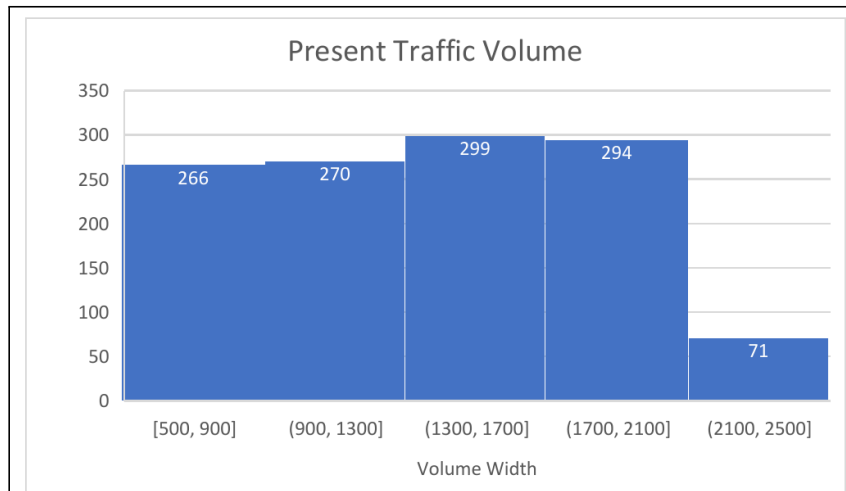


Figure 2: Frequency distribution (histogram) of present hourly traffic volume, by volume-width bin (as plotted in the original submission).

Conclusion: the **85th-percentile volume** is adopted as the estimated design volume instead of the 30th-hour value.

2.6 Step 3: Calculation of the 85th-Percentile Volume

To find the 85th percentile rank in a sorted (ascending) list of 1200 hourly values (50 days $\times$ 24 hours):

$$\text{Percentile Rank} = 0.85 \times 1200 = 1020^{\text{th}} \text{ value}$$

Table 5: Percentile ranks and corresponding design volumes.

Percentile	Position (th)	Design volume (PCU/hr/2-lane)
97.5th percentile	1164	2157
95th percentile	1140	2115
85th percentile	1020	1935

2.7 Step 4: Traffic Growth and Lane-Configuration Decision

Table 6: Design volume, v/c ratio, and LOS over 10 years for the 4-lane and 6-lane divided configurations, based on the 85th-percentile volume.

Year	1	2	3	4	5	6	7	8	9	10
Design volume	1935	2031.75	2133.34	2240.00	2352.00	2469.60	2593.09	2722.74	2858.88	3001.82
4-lane divided										
Capacity	4540	4540	4540	4540	4540	4540	4540	4540	4540	4540
v/c	0.4262	0.4475	0.4699	0.4934	0.5181	0.5440	0.5712	0.5997	0.6297	0.6612
LOS	C	C	C	C	D	D	D	D	D	D
6-lane divided (from year 4 onward)										
Capacity	–	–	–	6790	6790	6790	6790	6790	6790	6790
v/c	–	–	–	0.3300	0.3464	0.3637	0.3819	0.4010	0.4210	0.4421
LOS	–	–	–	C	C	C	C	C	C	C

Overall result. The 6-lane configuration, introduced from year 4 with a capacity of 6790 PCU per hour per lane, keeps the v/c ratio well below the LOS-D threshold, stabilising between approximately 0.33 and 0.44 over the remaining years. By expanding from 4 to 6 lanes starting in the present year, the road maintains LOS C throughout the initial years and avoids the drop to LOS D that would otherwise occur around year 4–5 under the 4-lane configuration. **This justifies a 4-to-6 lane capacity expansion** within the 10-year design horizon considered.

3 Assignment 3: Tangent Cross-Section of a Four-Lane Expressway in North-East India

Question. Develop the cross-section of a four-lane expressway for a tangent section. This expressway is located in the open country of the north-east region of India, where the annual rainfall is about 200 cm. Assume appropriate details from the relevant IRC code(s). Choose a suitable horizontal and vertical scale to represent the cross-section on graph paper.

3.1 Relevant Cross-Section Elements

Figure 3 reproduces the typical cross-section element taxonomy used to identify which elements are relevant to a tangent-section design. Of the full set of cross-section elements (pavement surface characteristics, cross-slope/camber, carriageway width, median/traffic separators, kerbs, shoulders, edge strips, road margins, and so on), only those highlighted in the original submission – width of carriageway, shoulders, median, road margins, roadway width, and right-of-way – were needed to develop this design.

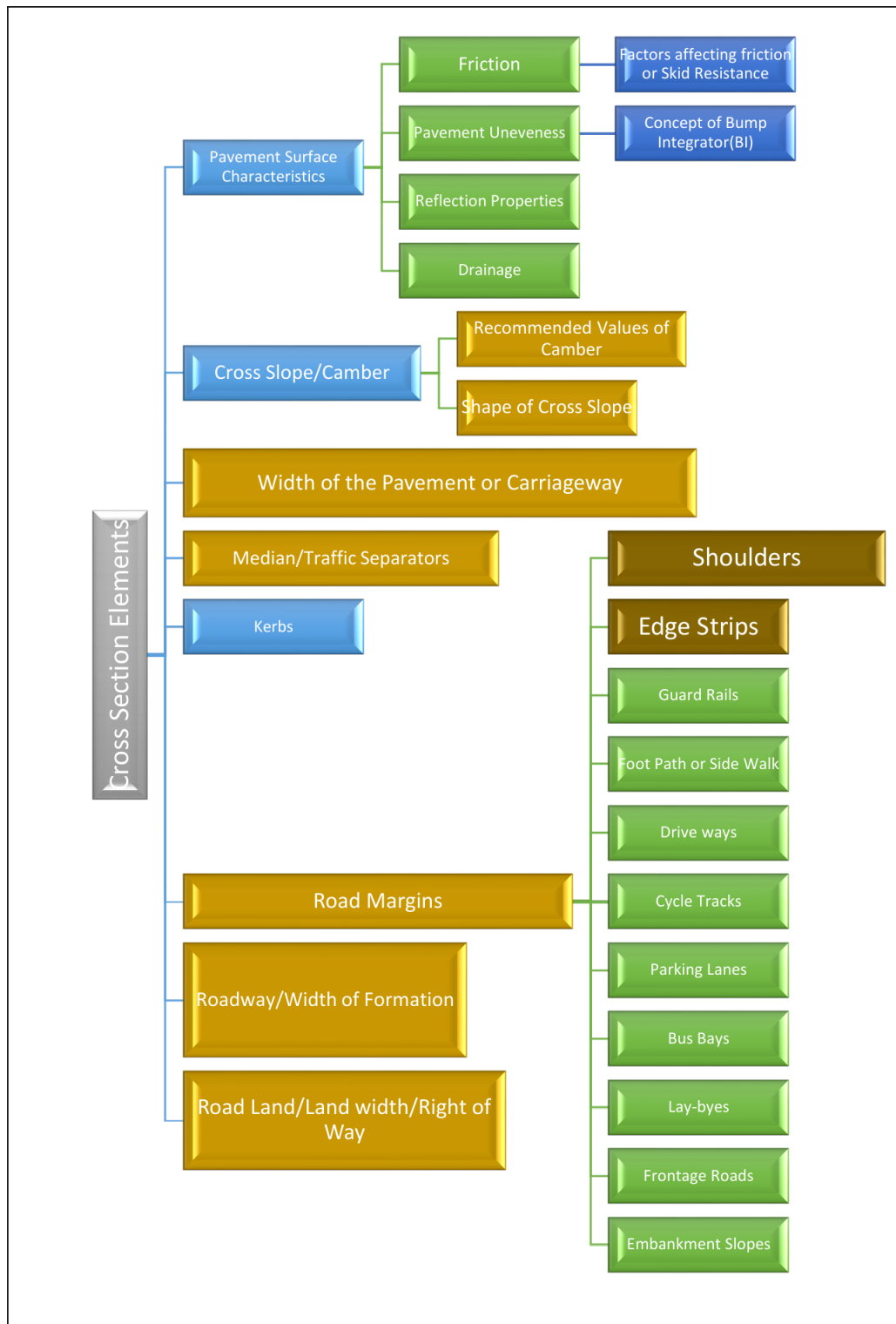


Figure 3: Typical highway cross-section elements considered when identifying which elements apply to a tangent-section design (as reproduced in the original submission).

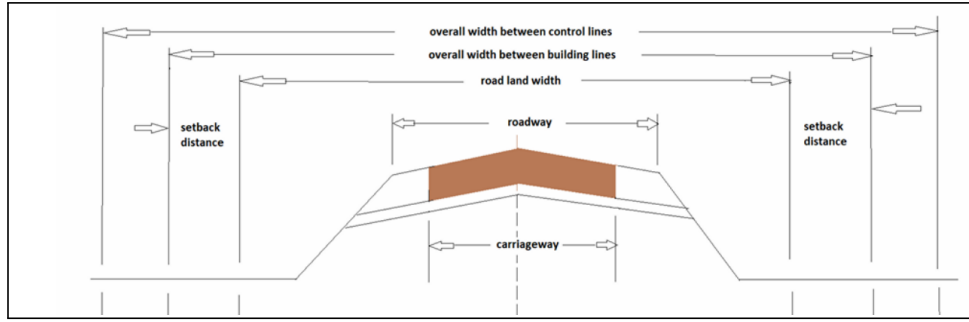


Figure 4: Generic annotated highway cross-section showing roadway, carriageway, setback distance, building line, and control line terminology used in the design (as reproduced in the original submission).

3.2 Design Considerations

1. **Location.** Open country in North-East India, with an annual rainfall of 200 cm.
2. **Road classification.** Expressway (4-lane).
3. **Pavement surface type.** IRC recommends a bituminous surface for high-speed roads.
4. **Lane width.** Per IRC guidelines for expressways in plain and rolling terrain, the lane width is 3.75 m per lane; for a four-lane highway, total carriageway width = $4 \times 3.5 \text{ m} = 14 \text{ m}$.
5. **Shoulder width.**
 - Paved shoulder: minimum 3 m on both sides.
 - Earthen shoulder: minimum 1.5 m on both sides.
 - Total shoulder width = $3 + 1.5 = 4.5 \text{ m}$.
6. **Median width.** A 10 m median is adopted (IRC/MoRTH minimum for expressways without a median barrier).
7. **Service lane.** A service lane of 5 m is adopted.
8. **Crossfall/camber.** For a bituminous surface in regions with heavy rainfall ($\geq 100 \text{ cm}$), the camber is **2.5%**.
9. **Side slopes.** For expressways, embankment side slopes of 4H:1V or flatter are recommended for moderate-height embankments.
10. **Right of way (RoW).** The RoW accommodates the roadway (carriageway, shoulders, median), side drains, service roads, and utilities. For a four-lane expressway in open country at a 100 km/h design speed, the RoW generally ranges up to **90 m** in plain/rolling terrain (per MoRTH guidelines for expressways).
11. **Building and control lines.** Per MoRTH guidelines for expressways in plain/rolling terrain, the overall width between building lines is 110 m and the overall width between

control lines is 130 m in open areas, with a 5 m distance between the building line and the RoW boundary in built-up areas.

Table 7: Summary of adopted cross-section parameters for the tangent section.

Parameter	Adopted value
Carriageway width (4 lanes)	14 m
Paved shoulder (each side)	3 m
Earthen shoulder (each side)	1.5 m
Total shoulder width (each side)	4.5 m
Median width	10 m
Service lane	5 m
Camber (bituminous, rainfall ≥ 100 cm)	2.5%
Embankment side slope	4H:1V (or flatter)
Right of way (plain/rolling terrain)	90 m
Width between building lines (open area)	110 m
Width between control lines (open area)	130 m

Note on the graph-paper deliverable. The question asked for the cross-section to be represented on graph paper at a suitable horizontal and vertical scale. The submitted PDF for this assignment ends at item 11 (Building and Control Lines) with the IRC/MoRTH reference tables reproduced above; it does not contain a separate scaled, dimensioned cross-section drawing on graph paper beyond the generic annotated diagram shown in Figure 4. This is reproduced here faithfully as submitted, without inventing a graph-paper drawing that is not present in the original assignment.

4 Assignment 4: Geometric Design and Safety Review of the H10 Intersection, IIT Bombay Campus

Group submission note: This assignment was completed jointly with Karshan M. Kanjariya (24D0322), Irfan Ali (24D0314), Menbere Aklilu (24D0315), and Fred Ssemwogerere (24D0326), as recorded on the original submission's title page. The content below reproduces the submitted group report.

Objective and scope. The objective of this study is to conduct a geometric design and safety evaluation of the H10 intersection within the IIT Bombay campus, grounded in the principles of intersection design taught in CE 773. The evaluation is based on field observations, photographic evidence, and intersection design guidelines, focusing on geometric elements such as sight distance, alignment, pedestrian and cyclist integration, and signage placement.

4.1 Location and Purpose of the Intersection

The selected intersection is located near Hostel 10 (H10) on the IIT Bombay campus, connecting the H14 hostel road, the H12 hostel approach, and a minor arm leading toward CCD and the KV School. It facilitates movement for hostel residents (pedestrians and cyclists), institutional traffic (autos, service vehicles), and access to key campus facilities (IDC, Hostels 10, 12, 14), experiencing moderate daily traffic as a T-intersection with a mixed user group.

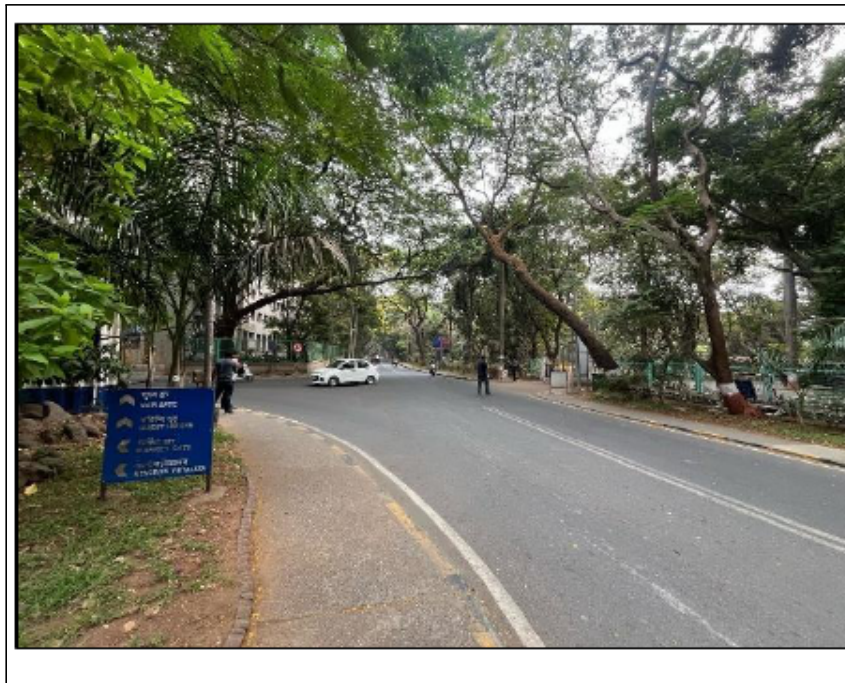


Figure 5: View from the H12 side toward the intersection, with visible signage and road curvature (Fig. 1 of the original submission).

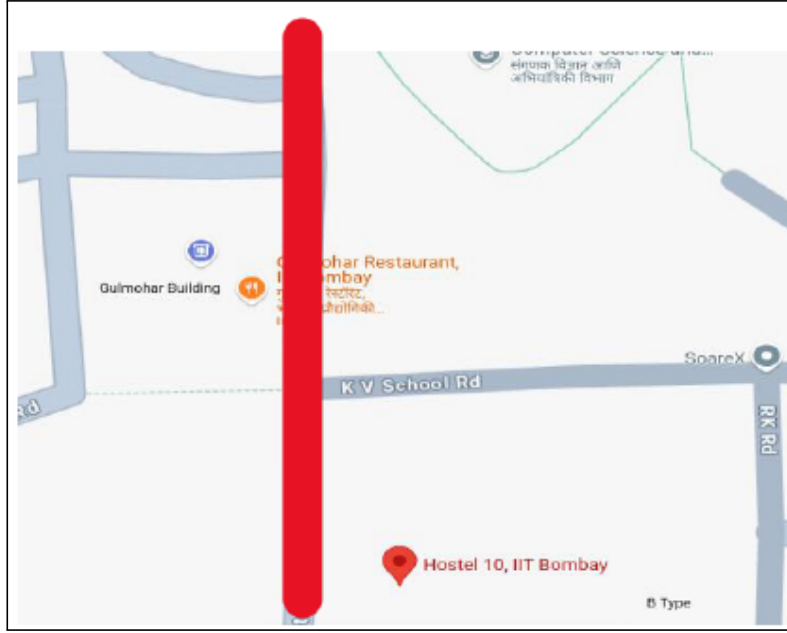


Figure 6: Google Maps aerial view of Hostel 10 and the surrounding intersection (Fig. 2 of the original submission).

4.2 Existing Layout and Traffic Movement

- **Intersection type:** 3-legged (T-intersection), near H10 and H12 hostels.
- **Traffic control features:** no yield or stop sign is provided; the intersection operates in an uncontrolled condition, relying on driver judgment and road-user negotiation.
- **Lane and movement characteristics:** each leg consists of two lanes with two-way traffic permitted; all turning movements are allowed, including right turns from the minor leg (H12) onto the major road and left turns into/from all approaches.
- **Traffic composition:** a heterogeneous mix of passenger vehicles (cars, two-wheelers, autorickshaws), non-motorized users (pedestrians and cyclists), and service/delivery vehicles (including light and heavy commercial vehicles accessing the SAC and hostels).
- **Geometric challenges from movement:** the absence of channelization and traffic control creates crossing conflicts between right-turning vehicles and oncoming traffic, merging/diverging conflicts at the T-junction without geometric separation, and pedestrian-vehicle confusion due to the lack of crosswalks and refuge islands.

4.3 Conflict Point Analysis

Based on the turning and through-movement possibilities at this three-legged unsignalized intersection, the following conflict points were identified.

Table 8: Conflict point count at the H10 intersection.

Conflict type	Count
Crossing conflicts	3
Merging conflicts	4
Diverging conflicts	4
Total conflicts	11

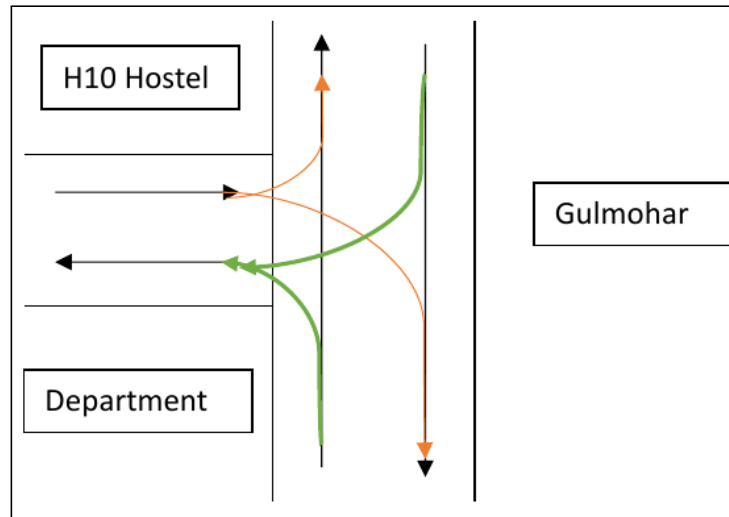


Figure 7: Schematic of traffic movements at the H10 intersection (H10 Hostel, Department, and Gulmohar arms), used to enumerate crossing, merging, and diverging conflict points.

Observation of conflict risk – obstructed sight triangle. Right-turning vehicles from the H12 approach face visibility issues due to vegetation or parked vehicles, increasing the likelihood of side-impact conflicts at crossing points.



Figure 8: Block Visibility: vegetation obstructing the sight triangle at the H12 approach (Fig. 3 of the original submission).

No pedestrian crossings. Without marked crossings or refuge islands, pedestrians cross informally, creating uncontrolled pedestrian-vehicle conflict zones that overlap with vehicle paths. **Lack of regulatory signs or markings.** The absence of Stop or Give Way signs and markings means drivers must guess priority, leading to unregulated merging and diverging behavior, especially during peak hostel hours.



Figure 9: No Zebra-Crossing near intersection (Fig. 4 of the original submission).

4.4 Skew Angle and Intersection Geometry

Based on the layout and road alignment visible in the field map and physical inspection, the angle of intersection formed at the H10 intersection is estimated to be approximately **85° to 90°**. The recommended angle for direct crossings is 75°–105° per standard design guidelines; the H10 intersection falls within this range and is therefore classified as a “Direct” intersection geometry.

4.5 Sight Triangle Evaluation

Intersection control type: Case A – No Control. There are no stop signs, yield signs, or signals on any approach; drivers must rely entirely on visual judgment to identify gaps and cross conflicting traffic streams.

4.6 Problem Identification

1. **Lack of dedicated space for non-motorized users.** No cycle lanes or shared pedestrian-bicycle pathways, leading to conflicts between vulnerable users and motorized vehicles.
2. **Inadequate and obstructed traffic signage.** Traffic signs are either too small, poorly placed, or hidden by vegetation, reducing their visibility and effectiveness.
3. **Unclear right-of-way due to missing regulatory elements.** No Stop or Give Way signs or pavement markings, resulting in driver confusion and unregulated movement.

4. **Non-standard and unsafe pedestrian infrastructure.** Uneven kerbs at bell mouths, non-uniform walkway heights, and speed breakers and footpaths at the same level.



Figure 10: Same level of speed breaker and footpath (Fig. 5 of the original submission).



Figure 11: Missing Pavement Marking: faded or missing pavement markings reduce lane definition and visual guidance, especially under low-light or wet conditions (Fig. 6 of the original submission).

5. **Sunken or damaged kerbs.**



Figure 12: Damaged Kerb (Fig. 7 of the original submission).

4.7 Geometric Design Evaluation

Table 9: Geometric design evaluation ratings.

Category	Rating
Uniformity	Needs Improvement
Conflict Minimization	Needs Improvement
Safety	Poor
Alignment & Profile	Acceptable

4.8 Conclusion

While the intersection geometry (angle and profile) is acceptable, the overall intersection design does not align well with core design principles: the lack of regulatory signage and markings undermines simplicity and uniformity; high conflict points and unmarked user paths raise safety concerns; and the absence of pedestrian and cyclist facilities makes the intersection non-inclusive. This makes a strong case for geometric redesign, particularly through channelization, visibility clearance, signage and markings, and dedicated paths for non-motorized transport (NMT).

4.9 Proposed Design Improvements

A. Non-motorized user facilities.

1. Dedicated cycle lanes / shared pedestrian-cyclist pathways along intersection arms, to reduce lateral conflicts between vulnerable users and motorized vehicles.
2. Marked pedestrian crosswalks (zebra crossings) on all legs to channel pedestrian movement.
3. Uniform pedestrian walkway height across approaches, to eliminate tripping risks.

4. Level kerbs, with standardized curb heights near turning zones for safer pedestrian crossing transitions.

B. Speed control and driver behavior management.

5. Speed breakers / rumble strips on all approaches at appropriate distances, to reduce speed and improve driver attention.
6. Rectify the oversized speed hump on the H12 arm, adjusting its height so the walkway remains elevated and the hump provides only mild deceleration.
7. Post speed-limit signboards (e.g., 25–30 km/h) using standard campus signage.

C. Signage, markings, and priority control.

8. Install regulatory signs (STOP / GIVE WAY) on the H12 approach and other minor arms to assign clear right-of-way priority.
9. Replace obscured or small traffic signs with more prominent retro-reflective signs, placed at standard eye level and cleared of vegetation.

D. Visibility and sight triangle enhancements.

12. Clear the sight triangle on the H12 arm by removing vegetation and visual obstructions within 25–30 m of the driver’s decision point on the minor arm.

E. Geometric redesign suggestions.

13. Repair or replace sunken median kerbs, using raised medians or plastic channelizers to clearly separate turning traffic and discourage corner-cutting.
14. Close the Canara Bank driveway entry, blocking vehicular entry from the bank side near the junction, which currently introduces unexpected diverging paths and increases merge conflicts.

Table 10: Summary of proposed improvement categories and key actions.

Category	Key action
Pedestrian/cyclist safety	Crosswalks, kerb levelling, shared pathways
Speed control	Rumble strips, proper hump design, signs
Traffic control	STOP/GIVE WAY signs and markings
Visibility	Sight-triangle clearance, sign upgrades
Geometric enhancements	Kerb repairs, channelization, driveway closure